ESSnet Big Data II

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https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata
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Workpackage WPA

Coordination and Communication

Deliverable A3: Intermediate Technical Report

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Executive Summary

The ESSnet Big Data II has just entered the second half of its lifetime. This report describes the progress of the project so far, by summarizing the results achieved, relating these results to what was required and specified in the grant agreement, and showing what resources have been used so far. Concerning the deliverables, by far most of those due by the end of January 2020 have actually been produced in their final version, after a review by the Review Board of the ESSnet. Two deliverables (E2 and K3) had to be rescheduled to 2020 for various reasons. All changes were agreed on by the Coordination Group of the ESSnet. The state of the milestones is very satisfactory overall. Concerning the resources used, these show generally a good match with the outputs.

1. Introduction

This deliverable of WPA[1] of the ESSnet Big Data II is the Intermediate Technical Report for the ESSnet as a whole. The report covers both the outputs of the project so far and the inputs (resources) used in the project. For both, the grant agreement is the main reference, but where there have been changes in the deliverables and milestones foreseen, or in the associated resources, this will be specified in this report. Links are provided to the project outputs on the wiki of the ESSnet. The Final Technical Report (deliverable A5) will include an overview of the contents of what the ESSnet will have produced in the end.

For the management of the project, meetings are regularly held by the Coordination Group (CG), in which the WP leaders plus the project manager from the side of Eurostat participate. From November 2018 to January 2020 a total of seventeen CG meetings were held. There are three types of CG meetings: full meetings (all WP leaders), Implementation Track meetings (WP leaders of WPA to WPF, plus WPK and WPL), and Pilots Track meetings (WP leaders of WPG to WPK, plus WPA and WPL). For all CG meetings minutes were written and approved. These minutes can also be accessed on the wiki. They are in fact a track record of the progress of the ESSnet. All changes mentioned in this report were agreed on in CG meetings and documented in the minutes.

As a rule, the deliverables of the ESSnet are reviewed by the Review Board (RB) of the ESSnet before they are delivered in their final form. The RB consists of independent professional statisticians from NSIs or academia. The RB is indispensable for the quality assurance of the ESSnet. At the start of the project, the RB consisted of two members. As one member changed job and there was a need anyway to extend the membership, given the high number and diversity of the deliverables, three new members were admitted in the

[1] Throughout this report, the term "Workpackage" is abbreviated to "WP". For a list and description of the twelve WPs (i.e., WPA to WPL), see the grant agreement.

second half of 2019[2]. As a consequence of the vacancies, there was some delay in the review process of deliverables, but this delay could be accommodated.

Concerning the inputs (resources) of the project used so far, the report gives information on what has been spent by each partner and each WP, in order to be able to assess whether this is reasonable compared to the progress made so far. The resources spent should neither be too high nor too low, given the current stage of the project.

Regarding the payments of ESSnet funds to the partners, this is not an issue for this intermediate report, as the grant agreement stipulates that 50% is paid after the start of the project, the rest being paid after its end. There were no intermediate payments foreseen. As far as the initial payments are concerned, the total amount was received by CBS (Netherlands) from Eurostat for distribution in accordance with the grant agreement and the initial budget. By now all but one of the partners have received their share (one partner experiencing some administrative delay in its request).

The structure of this report is simple. The next chapter (chapter two) gives a summary of the results so far. The subsequent chapter (chapter three) describes the realization of the deliverables of the project so far, including changes that were deemed necessary after the start of the project. In addition to the deliverables, milestones are also briefly looked at. Chapter four describes the resources spent so far and the necessary changes to the budget.

2. Summary of results so far

WPA Coordination and Communication

WPA covers general and financial management, coordination and administration of the ESSnet, in addition to communication and dissemination. The collaboration in the ESSnet went, overall, smoothly, and no major bottlenecks were encountered. As this report describes, all results of the ESSnet that were due by the end of January 2020 were realized, except for a limited number of cases (specified in this report) where all concerned agreed on a change in planning. Apart from the internal communication in the form of CG and other WebEx meetings, and internal meetings of the separate WPs, two physical meetings for the ESSnet as a whole were held in Vienna in December 2018 and December 2019, respectively. These meetings consisted of back-to-back meetings of the Implementation and Pilots Tracks. For internal and external communication, the wiki of the ESSnet proved to be very effective and indispensable.

[2] The RB is chaired by Lilli Japec (Statistics Sweden). The other members are now Giulio Barcaroli (Italy), Florian Keusch (Mannheim University) and Boriska Tóth (Statistics Norway).

WPB Online Job Vacancies

The interim technical report (deliverable B1) describes the progress of the WP in the first half of project execution. It summarises the activities that have taken place and provides information about the results and conclusions based on those activities. An important factor is the role of the European Centre for the Development of Vocational Training (CEDEFOP) as partner and likely future source of web-scraped online job advert data (OJA). The cooperation with CEDEFOP ensures that the data meets the needs of official statistics as far as possible and NSIs will be able to reduce their activities around data access and data handling and focus more on the challenges around further methodological development.

The Reference methodological framework (deliverable B2) describes at the conceptual level the complete statistical OJA data processing pipeline according to the four main phases: data ingestion, data pre-processing, information extraction and analysis, and production of statistical indicators. It addresses key conceptual, methodological, technical models and organisational principles and guidelines in using OJA data in official statistics. The idea is that such a framework would be used by statistical agencies as a reference guide when producing statistics using OJA data. The aim of this framework is therefore to:

- Provide a common language for various stakeholders
- Encourage adherence to common standards, specifications, methodologies
- Provide consistent methods for implementation
- Illustrate various implementation scenarios by identifying common components, processes and systems
- Provide a reference for statistical agencies and other parties to understand, discuss and compare OJA processing systems

Due to high complexity and heterogeneity of related OJA data pre-processing, this framework also discusses the role of a potential common OJA data collection and processing system (for the time being based on the OJA system developed by CEDEFOP) in the form of a future Web Intelligence Hub. The framework consists of two main parts, methodological considerations around OJAs and an ESS OJA reference production architecture. The latter takes into account developments related to the Big Data Reference Architecture and Layers (BREAL) framework, developed by WPF.

WPC Enterprise Characteristics

During the first year of the project, the following activities were completed:

(1) Elaborating of the ESS webscraping policy template (deliverable C1). The purpose of this policy is to ensure that webscraping activities of ESS members, namely Eurostat and the NSIs, are carried out transparently, consistently, ethically, respecting the acquis communautaire, as well as national legislation. The ESS webscraping template has been published on the wiki of the ESSnet. The template can be adapted to national requirements. The ESS webscraping policy template was presented at the ESS Task Force Big Data for

Official Statistics in October last year and it should be further discussed and potentially adopted as European guidelines.

(2) Elaborating of the Reference Methodological Framework version 1.0 (RMF 1.0). The RMF's purpose is to describe, at a conceptual level, a complete on-line based enterprise characteristics (OBEC) statistics processing pipeline across the four main big data life cycle phases: data acquisition, processing, modelling and interpretation, and producing statistical indicators (dissemination). The RMF 1.0 has been published on the wiki site as deliverable C2 and it is intended as a reference guide and template for NSIs within the ESS when setting up their own OBEC processing pipelines.

(3) The experimental statistics for 2019 developed in the context of WPC (deliverable C3) have been published on the wiki site, as besides tables with results for each of the WPC participant countries, methodological explanatory notes showing how these results were achieved were also published.

In 2020 the forthcoming activities are about elaborating RMF v.2, Starter Kit for the NSIs, production of experimental statistics for 2020 and filled in Quality template for experimental statistics on OBEC.

WPD Smart Energy

WPD aims to implement smart meter data for production of official statistics. During the project WPD will produce three statistical products: 1) electricity statistics of businesses, by sector; 2) electricity statistics of households; 3) identifying vacant or seasonally vacant dwellings. At the current stage the WP has developed a quality framework for electricity hub data that covers the input data, the process of producing statistics, and the statistical output. Also data validation rules have been developed that should aid NSIs to increase accuracy, reliability and efficiency during data transfer and during processing. A common methodology for geospatial linking and assessing the quality of linking has been developed. Prototypes for the three main statistical products are being tested. The methodology for finding electricity statistics of businesses and vacant dwellings is being fine-tuned and new classification algorithms are being tested.

WPE Tracking Ships

For each of the four main products, a first version of the process using AIS data has been developed:

(1) Develop statistics on the behaviour of fishing vessels: Using AIS, the first results on the activity of fishing fleet (time when a fishing vessel is out of the port) and the traffic of fishing fleet (movement of fishing vessels in fishing areas) in Poland have been derived. Several visuals have been created: map with the fishing ports and fishing areas, and the traffic intensity. The fishing fleet was derived using the EU fishing fleet register, applying the rules

of the Common Fisheries Policy. Next steps are to create experimental statistics and to test the solution for selected fishing ports in Greece and/or The Netherlands.

(2) Improve quality of statistics on port visits: A process workflow has been developed using AIS data to derive the port visits and port traffic of the Piraeus port in Greece. Also a visualisation (animation) has been created based on the AIS data, showing the traffic of the Piraeus port on March 30th, 2016.

(3) Improve statistics on Inland Waterway transport: Currently, statistics on Dutch Inland Waterways (IWW) are based on incomplete information. Information on certain ships, some parts of the Dutch inland waters and some goods carried is missing. Using AIS enables the WP to complete information on missing ships. Next step is to estimate the type of goods ships might have carried, based on the type of terminal.

(4) Improve statistics on air emissions and energy used: The first step of calculating the annual energy use and the air emissions from sea and coastal passenger and freight transport and fishery, is to derive all ships from Dutch shipping companies. Using text mining techniques, the WP tried to match the ships in the shipping list with the shipping companies in the business register. More than 60% of the ships are not matched, so more research is needed.

The products mentioned above were created using several big data platforms: local platforms, the BDTI (Big Data Test Infrastructure) environment provided by European Commission, and the UN global platform. The intermediate report will contain some paragraphs to share our experiences.

Finally, a first version of the WPE architecture was created using BREAL, as defined by WPF.

WPF Process and Architecture

WPF is working on the definition of a European reference architecture for big data, named BREAL (Big Data REference Architecture and Layers). BREAL serves the purpose of guiding big data investments by NSIs and helping the development of standardized solutions and services to be shared within the ESS and beyond.

The first achievements of WPF are related to a set of artefacts of the Business Layer of BREAL, namely: (i) a set of principles, (ii) a set of business functions, (iii) a description of the big data based production process called Big Data Life Cycle and (iv) a set of Actors and Stakeholders.

WPG Financial Transactions Data

The infrastructure and metadata of financial transactions data in the WPG countries have been investigated and described in deliverable 1. Also, official statistics that are promising candidates for being improved by financial transaction data, have been identified.

WPG countries have worked on accessing financial transactions data from different data providers, and some data sets have been obtained or will soon be obtained.

Different definitions of sharing economy have been discussed and one operational definition has been selected for the use in WPG.

A framework is being developed for deriving indicators on the sharing economy activity, where the input data to the indicators are tables based on financial transactions data. The overview of relevant financial transactions data tables in the WPG countries was based on a survey within WPG.

WPH Earth Observation

On the basis of satellite data from Sentinels in agriculture, the data have been pre-processed and prepared for image classifications using machine learning algorithms. Test results show that 20 single crops types can be identified with overall accuracy classification from 81% to 89%. The smallest accuracy was obtained for Decision Tree (DT) algorithm and the biggest for Artificial Neural Network (ANN). Due to the fact that the overall accuracy is affected by accuracy of each crop type recognition and the fact that cereals have similar characteristics leading to misclassification, the classes were re-defined. Cereals were divided into two group: spring and winter cereals, thereby increasing overall accuracy to 93% for ANN, 87% for DT. There is still some misclassification between fruit trees and fruit bushes plantations. In the case of monitoring of the off-season vegetation cover, the Random Forest (RF) was tested, but higher accuracy is expected and it prompted to extend the area of interest. Unfortunately, at this time aerial images cannot be used for crop recognition via machine learning, because they are recorded at an insufficient frequency (annually at low frequency i.e. annually and tri-annually, and even 10-yearly for the highest-resolution data) to constitute the time series needed for analysis.

The second thematic field of WPH is related to the urban areas. The task of delineation of urban areas based on OSO map (CS4) as well as Landsat images (CS5) works well. The computing time based on OSO map is around 30 minutes for one city, depending of the size of the city (it is slower for big cities). For the moment, the method was applied for the main 20 cities in France. It appears that the results are very close to the urban units. The concept of the urban unit is based on continuity of the built-up fabric. Every two buildings at a distance of less than 200m are considered contiguous. The delineation within cities boundaries of the areas which are open for public use is still under development. The task of delineation of urban areas based on Landsat has been performed using RF algorithm. Satisfying results caused the tests to be extended in area (more tiles) and in time (including older acquisitions from last years). Within research over urban areas the quality of housing is also under investigation. The EO data for this purpose has been collected and initially processed. The re-scale of the data is needed. Microcensus data has just been gathered and has not been processed yet.

The aim of the third thematic field of WPH, focusing on Land Cover (LC), is mainly improvement of existing systems for statistical production. In one case an initial comparison analysis between the Teruti database and the OSO product was carried out. Aim is to identify the differences in area by land cover class between the two methods and thus highlight the most different classes. The deeper analysis of differences at individual levels of Teruti points and OSO pixels is planned. Then some methods of detecting changes will be tested but depending on their implementation, their performances and more broadly their relevancies, some may ultimately not be retained. The second study related to land cover is focused to design and develop an automatic LC estimation system. The first automated LC maps were obtained.

In reference to the last subject, i.e. Update the INSPIRE Theme Statistical Units dataset and preventing forest fire, the preliminary results have been obtained concerning the identification of potential urban areas using COS2015 and CLC18 data.

WPI Mobile Networks Data

WPI aims at providing the construction of a production framework for the incorporation of mobile network data in many statistical domains. In this line, the Reference Methodological Framework proposed by Eurostat is taken as guidance, which WPI substantiates with concrete methodological proposals and prototype software. To avoid the extreme difficulties in getting access to this data source, a concurrent approach coping with different aspects is followed. The work is divided into eight deliverables (access, data simulator, methodology, IT, standards and metadata, quality, application on real data, visualization). The tasks on access and on the application on real data depend directly on agreements with MNOs and have been put off to the end of the project to incorporate as much up-to-date information as possible (e.g. the empirical assessment of the principles of access to privately held data). The data simulator follows a modular design so that its evolution is possible in order to incorporate different choices by the user (territory, population displacement pattern, network configuration,…). It is already available for closed populations with simple handover mechanisms and displacement patterns. The core of the methodology, heavily following a modular approach, comprises the geolocation of mobile devices in a reference grid, the computation of the distribution of the number of devices in this grid, and the computation of the distribution of the number of individuals. These methodological proposals are accompanied by prototype software tested using datasets from the simulator. Some R packages for the geolocation of devices implementing the proposed methodology have already been developed together with some preliminary visualization functions (including animated graphics). The computation of the number of devices is currently under study using these datasets. The estimation of the number of individuals will continue the hierarchical Bayesian modelling proposals initiated in the ESSnet Big Data I. The work on standards/metadata and quality will start in 2020 and will follow developments from WPF and WPK in the quest for standardization.

WPJ Innovative Tourism Statistics

WPJ implemented a number of tasks through the first year of the project. An inventory of external data sources and a selection of webscraped portals related to accommodation booking services broken down to those with global and local reach was made. Based on the experience gained during the webscraping process, a description of the open source libraries and implemented technologies was prepared, that later was used to create a universal tool (Visual Modeler) that facilitates downloading data from dynamic websites. Then, using selected methods, an attempt was made to uniquely identify accommodation establishments that were found in webscraped databases, administrative registers and survey frames for tourist accommodation establishments. Pairing of establishments was done by comparing address data as well as using a more advanced approach, namely a method based on the use of geographical coordinates of the location of the establishments using solutions of Here Maps API. This allows to combine data from different sources faster by automatically converting address data structures to a uniform format.

WPK Methodology and Quality

The aim of WPK is to consolidate knowledge gained in this ESSnet with respect to methodology and quality when using big data in the statistical production process and combine it with the insights from the previous ESSnet Big Data I. The key deliverables with respect to quality were the draft versions of a quality guideline and a quality reporting template. Upcoming work will focus on methodology and improvements of the quality related documents. The typification of big data projects is another strain of work in this WP and is somehow linking the first two topics. As the input of all WPs from this ESSnet project is of high importance to WPK, the kick-off and intermediate workshop were organized and facilitated by members from WPK.

WPL Preparing Smart Statistics

In WPL, a wide variety of topics concerning the subject IoT have been examined and many promising data sources have been identified. These refer in particular to smart farming, smart cities, smart devices and smart traffic. The results show that it might be rewarding to dig deeper in follow-up studies. However, generally the data landscape is still in a development state. Data access and availability are often an issue. Even though merging of sensor data with geolocalized official data has been tested successfully and some data sources are already used for estimates and have been published as research papers, the data format and the transferability to the ESS still has to be examined in many cases. Also, data protection is an important subject to take care of by official statistics, but there are quite promising trends leading to open data, which is usually anonymized and aggregated. Furthermore, the confidentiality of data could be protected by using new technologies and procedures such as cryptography technologies or blockchains. Certainly, these technologies

have to be further investigated. In addition, the quality of data gathered by smart systems and their representativity should be examined carefully as well.

Altogether the use of data generated by IoT tools offers many possibilities to enrich official statistics, but both using new data sources and adopting/changing the statistical production processes of the NSIs to involve those have to be explored in further studies.

3. Project output

3.1 Project deliverables

In this section all deliverables due by the end of January 2020 according to the grant agreement are listed for each of the WPs. If they were realized in 2019 or the first month of 2020 in their final version, this is simply noted. If they have not been realized yet, or have not been finalized yet, this is not only noted, but also explained. An outlook for the remainder of the project is given as well. Minor delays that occurred before January 2020 for deliverables that have nevertheless been produced by now are not mentioned.

deliverables of WPA

Deliverables due by January 2020:

- A1 ESSnet communication and collaboration platform running
- A2 ESSnet Big Data website online
- A3 Intermediate technical report

Realized:

Both A1 and A2 were realized, see wiki. A3 is this report.

Outlook:

The remaining deliverables (A4 and A5) are on schedule. Preparations for the BDES 2020 conference have started already. The conference will take place in Warsaw, Poland, on 25 and 26 of November, 2020. On the 27th of November there will be a final ESSnet meeting for ESSnet participants to the BDES 2020 conference and a back-to-back CG meeting.

deliverables of WPB

Deliverables due by January 2020:

- B1 Interim technical report
- B2 Methodological Framework V.1

Realized:

Both B1 (see link) and B2 (see link) were realized.

Outlook:

The remaining deliverables (B3, B4 and B5) are on schedule.

deliverables of WPC

Deliverables due by January 2020:

- C1 ESS web-scraping policies (report)
- C2 Reference Methodological Framework for processing online based enterprise characteristics (OBEC) data for Official Statistics V.1
- C3 Experimental Statistics 2019, including reference metadata

Realized:

C1 (see link), C2 (see link) as well as C3 (see link) were realized.

Outlook:

The remaining deliverables (C4 to C8) are on schedule. The proposed web-scraping policy will be considered for adoption by the ESS, it is now being discussed by the appropriate groups in the ESS governance structure.

deliverables of WPD

Deliverables due by January 2020:

- D1 Technical report V.1
- D2 Technical report V.2

Realized:

Both D1 (see link) and D2 (see link) were realized.

Outlook:

The remaining deliverable (D3) is on schedule.

deliverables of WPE

Deliverables due by January 2020:

- E1 Report on the results of the definition process
- E2 Report on implementation requirements of relevant statistical production processes, quality framework, required metadata and a risk plan

Realized:

E1 has been realized (see link). E2 is not ready. As agreed by the CG, E2 will be combined with E3 into one report. E3 concerns the interim technical report on conditions and guidelines for using the data, the methodology and the procedures. The rescheduling was considered the most practical solution to issues of data acquisition and platform availability, which led to using the UN Global Platform and the AIS data available there (see also the minutes of the CG meetings).

Outlook:

The remaining deliverable (E4) is on schedule.

deliverables of WPF

Deliverable due by January 2020:

- F1 BREAL: Big Data REference Architecture and Layers, Business Layer

Realized:

F1 was realized (see link).

Outlook:

The remaining deliverable (F2) is on schedule.

deliverables of WPG

Deliverables due by January 2020:

- G1 Report on a selection of the tasks of WPG with country-specific information
- G2 Conclusion on studies and plan for the second year
- G3 Closing the selection of studies (concerning statistics, financial transaction data type, country)

Realized:

G1 (see link), G2 (see link) as well as G3 (see link) were realized. Concerning G1, the WP encountered difficulties in obtaining data on Shared Economy Platforms. This led to changes in scope of the deliverable, as discussed in several CG meetings and agreed on (see minutes), as well as a redistribution of tasks of the partners of the WP.

Outlook:

The remaining deliverable (G4) is on schedule.

deliverables of WPH

Deliverable due by January 2020:

- H1 Interim technical report concerning the Reference Methodological Framework and the conditions for using the data, the methodology and the procedures to be used for producing statistics

Realized:

H1 was realized (see link).

Outlook:

The remaining deliverables (H2 and H3) are on schedule.

deliverables of WPI

Deliverable due by January 2020:

- I2 A simulator for network event data

Realized:

I2 was realized (see link). It includes an installation video and a reference guide.

Outlook:

The deliverable I2 may be used for course purposes. The remaining deliverables (I1 and I3 to I8), however, which were and remain due in 2020, had to be rescheduled, due to substantial difficulties and delays in getting data access. Nevertheless, the aims of the WP remain the same. A new schedule for the remaining deliverables was agreed on during the intermediate meeting for the Pilots Track in December in Vienna (see the review planning). The most critical deliverable is I7, concerning experimental results with mobile phone data.

deliverables of WPJ

Deliverables due by January 2020:

- J1 ESSnet methods for webscraping, data processing and analyses
- J2 Interim technical report showing the preliminary results and a general description of the methods used

Realized:

Both J1 (see link) and J2 (see link) were realized.

Outlook:

The remaining deliverables (J3, J4 and J5) are on schedule.

deliverables of WPK

Deliverables due by January 2020:

- K1 First draft of quality guidelines
- K2 Updated literature overview with outside sources and additional input from the ESSnet
- K3 Revised version of the quality guidelines
- K4 Quality report template draft

Realized:

K1 (see link), K2 (see link) as well as K4 (see link) were realized. However, within 2019, K1 was produced much later than originally foreseen, due to the workload involved, which was much higher than anticipated. As K3 depends on input from K1, it was agreed by the CG to reschedule this deliverable to M17 (see CG minutes). There are no consequences for other deliverables.

Outlook:

The remaining deliverables (K5 to K12) are on schedule.

deliverables of WPL

Deliverables due by January 2020:

- L1 Description of findings regarding task 1, Smart Farming
- L2 Description of findings regarding task 2, The use of IoT for Smart Cities
- L3 Description of findings regarding task 3, Smart Devices
- L4 Description of findings regarding task 4, Smart Traffic

Realized:

L1 (see link), L2 (see link), L3 (see link) as well as L4 (see link) were all realized. In fact, WPL is the only WP that was planned to end in 2019, all its deliverables having M12 as scheduled deadline. All deliverables were produced in time, and the RB took an extra effort to facilitate this.

Outlook:

There are no deliverables remaining.

3.2 Project milestones

With one exception, all milestones are reports on physical meetings. The exception is the last milestone of WPL, which is its final report. This was produced on schedule (see link).

Also with one exception, all physical meetings planned for 2018 and 2019 took actually place in that period. The exception is the third physical meeting of WPI, which was postponed to February 2020 for reasons related to the rescheduling of deliverables described in chapter three.

For all physical meetings that took place in 2018 and 2019, minutes are available on the wiki (see WP meetings on this page). This includes the two intermediate meetings organized by WPF and WPK in December 2019 in Vienna for the Implementation Track and the Pilots Track, respectively. The minutes of these events also provide a good overview of the state of affairs of the ESSnet.

4. Project resources

4.1 Resources spent

The budget for the ESSnet was specified at a detailed level in the grant agreement. The extent to which the budget is being spent during the project is regularly monitored for project management reasons. At any time sufficient resources must be left to realize the remaining deliverables. Moreover, the use of the budget is an indication of what has been invested so far in the deliverables; if this is much lower than foreseen, that could signal a lack of progress. This would necessitate an investigation of the causes and possibly measures to catch up. The monitoring was done in five rounds, their timing tuned to CG meetings. The last round took place in December 2019. The information gathered pertains to the cumulative number of person/days and all financial resources spent since the beginning of the project (i.e., November 2018) per partner, using a template similar to the one of the original budget.

A summary of the latest results is given in the first two annexes to this report. The number of person/days and direct costs for each WP per country over the period 01-11-2018 to 01-12-2019 is shown in Annex 1. In order to be able to compare this to what could reasonably be foreseen at this point of the project, relative figures are provided in Annex 2. Before discussing the findings, the annexes are briefly explained.

Explanation of Annex 1

As no working days were budgeted in the grant for the NSIs of Ireland, Switzerland and the UK, these are not included in the table.

The overview contains the resources spent from the beginning of the project until 1 December 2019, for all other partners except for SSB (Norway), GUS (Poland) and STAT (Austria), whose figures are until 1 November 2019.

Whereas personnel costs are complete for all partners, at this moment not all travel and subsistence costs have already been included for each face-to-face meeting. Depending on the financial administration of the partner, there may be time lags in the registration of these costs.

Explanation of Annex 2

This overview gives a summary of the realization so far of the personnel costs in proportion to what might be expected at the level of the partner as well as per WP. The realization figures are based on those of Annex 1, and the same conditions apply. The proportional budget is based on the number of months since the start of the project relative to the total duration of the project. Where WPs end before the end date of the ESSnet, this has been taken into account. For instance, WPL ended, according to planning, in M12, and for this WP the proportional budget was set at 100%. The proportional budget does not take into account the timeline of the deliverables within WPs and their internal planning.

Findings

For the project as a whole, the realization of days and associated budget is 12 percent less than the proportional figures. Taking into account the fact that the grant was signed after the actual start of the project and the fact that all WPs (except for WPL) have relatively much work to do when the last deliverables are due before the project's end, this is not only very plausible overall, but also healthy. In terms of resources the project is on full steam, but still has some reserves for demanding times to come.

At the level of the individual partners, there are some differences. Five partners (of the twenty-five) have spent more days than what was to be expected so far (these cases are marked in the table), but there may be specific reasons for this and it is nowhere extraordinary. Moreover, the partners are aware of the fact that in the end, they may not be reimbursed for days that have not been budgeted.

At the level of the individual WPs, there are also some differences. The most important ones with underspending are WPA and WPD. WPA will catch up at the end of the project because dissemination (and thus the final conference BDES 2020) is part of this WP. Some of the partners of WPD temporarily lacked (human) resources, which influenced the days spent. WPD is expected to catch up later on. For three WPs (WPE, WPG and WPI) there is some underspending that is related to the rescheduling of deliverables (see chapter two).

The other WPs are in line with expectations. Especially for WPF and WPK substantial work will be done in the second half of the project because of the coordination of the work between and within the WPs.

4.2 Changes to the budget

As already mentioned in the grant agreement, there are bound to be unforeseen changes in any research project that lasts more than two years and has as many as 28 partners. As the budget is rather detailed, specifying for instance the grades and working days of all staff involved and all travel costs per partner per meeting, the budget realization at a detailed level will differ in a large number of instances from the original budget. Such changes will be dealt with in the final report after the end of the project, provided that these changes do not change the relevant total budget amounts concerned.

In particular, the participation to meetings differs sometimes from what was originally foreseen, for instance due to the location or agenda of meetings, staff availability, or specific needs related to particular WP activities. In some cases the schedule of meetings themselves has been changed, for instance allowing for back-to-back meetings. In such cases the changes that are needed are coordinated by the ESSnet coordinator with the Eurostat project officer. Care is taken that the changes to meetings, including participation, are kept within the boundaries of the budget for travel and meetings. An example is the change to the third internal meeting for WPI, mentioned in chapter two.

The changes in membership of the Review Board are another example where the final report will contain changes to the original budget, as the countries of the new members were not known in advance. Again, the changes will not lead to a budget increase for the Review Board as a whole.

There are two changes to the estimated budget that are worth specifying here, as they concern a transfer among partners. Again, these were coordinated with all concerned, and they will be implemented in the final report.

<u>Budget transfer 1:</u>

Reallocate travel and subsistence costs from Bulgaria and The Netherlands.

The WPF meeting in Rome was in May 2019. The participant from Bulgaria couldn't reserve a plane and hotel ticket due to their specific situation. The BNSI was in the middle of a public procurement process about logistics and the BNSI didn't have a signed contract with a logistics company at the time. Therefore they asked Statistics Netherlands to buy the tickets. When the final payment is done by Eurostat these costs will be settled between the two partners.

The total amount of the settlement is approximately € 900.00.

<u>Budget transfer 2:</u>

Redistribution of budget within WPG.

Within WPG there has been a shift of the work load from all partners to Istat (Italy), resulting from the redistribution of tasks explained in chapter two. As a consequence, the number of days per partner had to be changed. The redistribution is budget neutral within WPG. The detailed information is given in Annex 3.

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Annex 1: Resources spent on personnel and direct costs

	Essnet Big Data 2018-2020	2019-12-30	WPA: coordination and communication		WPB: online job vacancies		WPC: enterprise characteristics		WPD: smart energy		WPE: tracking ships		WPF: process and architecture		WPG: financial transactions data		WPH: earth observation		WPI: mobile networks data		WPJ: innovative tourism statistics		WPK: methodology and quality		WPL: preparing smart statistics								
	PERSONEL COSTS	Realization																															
	1,3		days	total	days	total	days	total	days	total	days	total	days	total	days	total	days	total	days	total	days	total	days	total	days	total		total	travel	other	100%	days	
1	Statistics Netherlands	CBS	141	91874			88	43885			139	67721	20	12422			44	21077	12	5502	41	19297	20	12860	31	20887		295525	7034	3075	305633	535	CBS
2	Statistics Belgium	SB	41	22773													5	2777										25551	611	0	26161	46	SB
3	Bulgarian National Statistical Institute	BNSI			146	8456	239	11747					133	5805	89	7283					93	5535			60	4204		43031	7101	0	50132	760	BNSI
4	Statistisches Bundesamt	DESTATIS			126	48345	6	2558					11	4203	27	9244	32	12543	4	1438					71	27450		105782	4356	18000	128138	275	DESTATIS
5	Statistics Denmark	DS							34	13641			6	3413														17054	1778	0	18832	40	DS
6	Statistics Estonia	EE							134	34577			48	12386					7	1806								48769	1868	0	50637	189	EE
7	Hellenic Statistical Authority	ELSTAT									116	21455									45	7423						28878	3288	0	32166	161	ELSTAT
8	National Institute Of Statistics	ISTAT			10	3044	47	17862					39	19117	59	24883	35	12257	70	21326	29	9597	29	11017	54	19873		138975	7273	0	146248	371	ISTAT
9	Statistics Norway	SSB	5	3490					18	9355					134	72019									12	6570		91433	3089	0	94523	169	SSB
10	Central Statistical Office Of Poland	GUS	0	0	47	4895	125	10932			47	4389					141	17199			190	19497	39	4154	50	9333		70398	5456	0	75854	639	GUS
11	Instituto Nacional De Estatistica	INE (PT)											12	2704	9	2505	37	10384			152	17796	6	1296	40	9346		44031	2689	0	46720	256	INE (PT)
12	National Institute Of Statistics of Romania	INS			38	3058													63	8007								11064	2059	0	13123	101	INS
13	Statistical Office of the Republic of Slovenia	SURS			74	17498									51	8645												26143	3149	0	29292	125	SURS
14	Instituto Nacional de Estadistica	INE (ES)																	102	33944			15	5117				39061	1567	0	40628	117	INE (ES)
15	Statistics Sweden	SCB	0	0	14	6909			16	10603																		17513	275	0	17788	30	SCB
16	Statistik Austria	STAT					46	26120															53	21975	0	0		48095	3072	0	51167	99	STAT
17	Statistical Office of Slovak Republic	SOSR																			41	7118						7118	391	0	7509	41	SOSR
18	Statistics Finland	SF	6	3076			35	13915																	29	12506		29497	5351	0	34848	70	SF
19	Natural Resources Institute Finland	LUKE															53	18514										18514	1186	0	19700	53	LUKE
20	Institut National De La Statistique Et Des Etudes Economiques	INSEE											20	16259			21	14081	41	26758					29	19173		76270	4661	0	80931	111	INSEE
21	Ministery Of Labour - France	DARES			51	26089																						26089	1503	0	27592	51	DARES
22	Ministry of Agriculture - France	SSP															17	10564										10564	2005	0	12569	17	SSP
23	Office for Statistics Berlin-Brandenburg - Germany	Berlin																							14	6797		6797	700	0	7498	14	Berlin
24	Hessen Statistical Office - Germany	Hesse																			31	12079						12079	637	0	12716	31	Hessen
25	Statistics Lithuania	LT			12	1613																						1613	1799	0	3412	12	LT
	TOTAL PERS. COSTS (incl. 30 %		192	121213	517	119907	586	127019	202	68176	302	93565	287	76309	369	124580	386	119395	298	98781	623	98342	163	56419	390	136139		1239845	72898	21075	1333817	4313	
			number	total	number	total	number	total	number	total	number	total	number	total	number	total	number	total	number	total	number	total	number	total	number	total							
	TRAVEL (note travel estimate is simplified in		0	0	0	7040	0	2244	0	1751	0	3415	0	16511	0	4757	0	2896	0	2741	0	2585	0	14697	0	14261		72897			72897		
	dissemination poland			0																													
	Kick-off meetings													11854										11290		6487							
	Mid/final-term meetings													1274										548		7773							
	internal/combined meetings					7040		2244		1751		3415	0	3383	0	4757	0	2896	0	2741	0	2585	0	2860	0	0							
	Direct costs of contracting			0		0		0		0		0		0		0		0		0		0		0		0		0			0		
	Subcontracting																	0								0							
	Other direct costs			3075		8000		0		0		0		0		0		10000		0		0		0		0		21075			21075		
	Meeting costs			3075		0																											
	dissemination workshop poland (100 people)			0																													
	Technical support media wiki			0																													
	Purchase of data and reports					8000												10000		0													
																						TOTAL COSTS :						1 333 817			1333817		

Annex 2: Resources spent by partner and WP, relative to the budget

	ESSnet BDII realization 2019-12-30			Realization						Budget			Total Estimated Costs			Proportional Budget	Realization		Proportional Budget	Realization		Proportional Budget	Realization	Work Package	Work Package
																€	€		days	days		days	days		%
				total pers.costs (incl. 30% indirect)	travel	other	100%	days		total pers.costs incl. 30% indirect)	travel	other	100%	days		total pers.costs (incl. 30% indirect) 2019-11-30	total pers.costs (incl. 30% indirect) 2019-11-30		Nov 18 - Nov19	Nov 18 - Nov 19		Per Work Package Nov 18 - Nov 19	Per Work Package Nov 18 - Nov 19		
1	Statistics Netherlands	CBS		295525	7034	3075	305633	535		549656	32560	3500	585716	1002		300132	295525		548	535		241	192	WPA	-20%
2	Statistics Belgium	SB		25551	611	0	26161	46		52768	7560	6260	66588	95		27045	25551		49	46					
3	Bulgarian National Statistical Institute	BNSI		43031	7101	0	50132	760		81002	13680	7320	102002	1471		45851	43031		823	760		571	517	WPB	-9%
4	Statistisches Bundesamt	DESTATIS		105782	4356	18000	128138	275		166106	15780	18000	199886	431		102174	105782		264	275					
5	Statistics Denmark	DS		17054	1778	0	18832	40		75446	5030	0	80476	175		41893	17054		98	40		604	586	WPC	-3%
6	Statistics Estonia	EE		48769	1868	0	50637	189		93734	9480	0	103214	405		52028	48769		225	189					
7	Hellenic Statistical Authority	ELSTAT		28878	3288	0	32166	161		53515	6550	0	60065	300		28987	28878		163	161		348	202	WPD	-42%
8	National Institute Of Statistics	ISTAT		138975	7273	0	146248	371		265207	19310	0	284517	702		149284	138975		395	371					
9	Statistics Norway	SSB		91433	3089	0	94523	169		157986	11250	0	169236	294		92292	91433		172	169		375	302	WPE	-19%
10	Central Statistical Office Of Poland	GUS		70398	5456	0	75854	639		160018	17440	11692	189150	1529		92717	70398		869	639					
11	Instituto Nacional De Estatistica	INE (PT)		44031	2689	0	46720	256		106193	12090	0	118283	448		62597	44031		264	256		342	287	WPF	-16%
12	National Institute Of Statistics of Romania	INS		11064	2059	0	13123	101		23621	6710	0	30331	234		12795	11064		127	101					
13	Statistical Office of the Republic of Slovenia	SURS		26143	3149	0	29292	125		45076	9480	150	54706	279		24416	26143		151	125		461	369	WPG	-20%
14	Instituto Nacional de Estadistica	INE (ES)		39061	1567	0	40628	117		60161	7960	3200	71321	180		32370	39061		97	117					
15	Statistics Sweden	SCB		17513	275	0	17788	30		95000	6620	0	101620	160		51986	17513		88	30		421	386	WPH	-8%
16	Statistik Austria	STAT		48095	3072	0	51167	99		81524	11090	0	92614	150		47626	48095		86	99					
17	Statistical Office of Slovak Republic	SOSR		7118	391	0	7509	41		20028	3690	0	23718	107		10849	7118		58	41		359	298	WPI	-17%
18	Statistics Finland	SF		29497	5351	0	34848	70		59406	6730	0	66136	144		37517	29497		92	70					
19	Natural Resources Institute Finland	LUKE		18514	1186	0	19700	53		18176	3690	20000	41866	52		11252	18514		32	53		580	623	WPJ	7%
20	Institut National De La Statistique Et Des Etudes Economiques	INSEE		76270	4661	0	80931	111		121777	9410	0	131187	182		78888	76270		118	111					
21	Ministery Of Labour - France	DARES		26089	1503	0	27592	51		72726	3690	0	76416	143		39393	26089		77	51		177	163	WPK	-8%
22	Ministry of Agriculture - France	SSP		10564	2005	0	12569	17		7457	3690	24856	36003	12		4616	10564		7	17					
23	Office for Statistics Berlin-Brandenburg - Germany	Berlin		6797	700	0	7498	14		7372	2350	0	9722	15		7372	6797		15	14		399	390	WPL	-2%
24	Hessen Statistical Office - Germany	Hesse		12079	637	0	12716	31		33120	3690	0	36810	85		17940	12079		46	31					
25	Statistics Lithuania	LT		1613	1799	0	3412	12		3616	3690	0	7306	26		1959	1613		14	12					
	TOTAL COSTS			1239845	72898	21075	1333817	4313		2410691	233220	94978	2738889	8620		1373977	1239845		4878	4313		4878	4315		-12%

Annex 3: Budget redistribution within WPG

Country/function	unit costs grade	Nr of persons	Days	Unit costs	Total per line	Total per country	Reduction %	Reduction in €	Total minus reduction	30% indirect pers cost	Total J and K	Reimbursement 90% of colomn L	Original budgeted reimbursement	Decrease reimbursement
WPG ONLY!!!								Increase Italy	Increase Italy	30%		90%	colomn G *130%*90%	Increase Italy
Bulgaria													130%	
State expert	14	1	60	43,04	2.582,40 €									
Director	8	1	44	81,22	3.573,68 €									
Head of Department	12	1	26	65,74	1.709,24 €									
State expert	14	1	21	43,04	903,84 €									
Senior expert	20	1	21	29,63	622,23 €									
Chief Expert	17	1	66	39,34	2.596,44 €									
						11.987,83 €	**12%**	**1.438,54 €**	10.549,29 €	3.164,79 €	13.714,08 €	12.342,67 €	14.025,76 €	€ 1.683,09
Germany														
Higher service	A13 hD	1	51	263,37	13.431,87 €									
						13.431,87 €	**4%**	**537,27 €**	12.894,60 €	3.868,38 €	16.762,97 €	15.086,68 €	15.715,29 €	€ 628,61
Norway														
Director	C	1	3	558,00	1.674,00 €									
Research professor	E	1	11	538,00	5.918,00 €									
Senior Adviser	H	1	126	402,00	50.652,00 €									
Advisor	I	1	16	326,00	5.216,00 €									
						63.460,00 €	**8%**	**5.076,80 €**	58.383,20 €	17.514,96 €	75.898,16 €	68.308,34 €	74.248,20 €	€ 5.939,86
Portugal														
Methodologist	32	1	34	214,09	7.279,06 €									
Statistical researcher	28	1	81	179,73	14.558,13 €									
						21.837,19 €	**8%**	**1.746,98 €**	20.090,21 €	6.027,06 €	26.117,28 €	23.505,55 €	25.549,51 €	€ 2.043,96
Slovenia														
Counselor III A (Statistician)	28	1	38	96,06	3.650,28 €									
Senior Counselor I B (Senior Statistician	42	1	30	167,20	5.016,00 €									
Secretary (Head of Unit)	51	1	3	246,62	739,86 €									
Secretary (Senior Statistician)	49	1	5	215,24	1.076,20 €									
Counselor III D (Statistician)	28	1	39	96,06	3.746,34 €									
						14.228,68 €	**8%**	**1.138,29 €**	13.090,39 €	3.927,12 €	17.017,50 €	15.315,75 €	16.647,56 €	€ 1.331,80
					Total	**124.945,57 €**		**9.937,88 €**	115.007,69 €					
Italy														
Senior statistician	2	1	18	354,00	6.372,00 €	**28,07 days extra work**								
Senior statistician	2	1	18	354,00	6.372,00 €									
Senior statistician	2	1	18	354,00	6.372,00 €									
Statistician	3	1	18	251,33	4.523,94 €									
Statistician	3	1	18	251,33	4.523,94 €									
Statistician	3	1	18	251,33	4.523,94 €									
Statistician	3	1	17	251,33	4.272,61 €									
Statistician	3	1	17	251,33	4.272,61 €									
Statistician	5	1	17	222,32	3.779,44 €									
Statistician	5	1	17	222,32	3.779,44 €									
						48.791,92 €							57.086,55 €	€ 11.627,32
					Extra work		**increase**	**9.937,88 €**	58.729,80 €	17.618,94 €	76.348,75 €	68.713,87 €		
				Total WPG Budgeted		**173.737,49 €**			**173.737,49 €**			**203.272,86 €**	**203.272,86 €**	**0,00 €**